SOCIAL SPIDERS

Group 16

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OBJECTIVES

1. Intro & Motivation
2. Related Work
3. Stakeholders
4. System Overview
5. Impact
6. Testing
7. Conclusion & Future Work

1.

Introduction & Motivation

Overview of our project and key motivations

Dylan

Motivation

- Social media allow children aged 13+ to post
  → Many will have child bank accounts or debit cards (NatWest 2026).

- Up to 95% of youth ages 13–17 report using a social media platform (OSG 2023).

- Social media encourages users to share updates about their lives.
  → Attackers use **social media** *profiles* and **public information** to facilitate attacks (Equifax UK 2016).

We want to empower the next generation to view their personal data as their most valuable financial asset, shifting them from passive targets to active guardians of their own economic future.

2.

Related Work

Similar solutions targeting similar problems

Related Works

Acorns Early (Acorns, 2026)	Aura (Aura, 2026)	Internet Matters (Internet Matters, 2026)
Financial Literacy	Social Monitoring	Digital Safety
Designed for ages 6-18	Any age: targeted at families	Resources for all ages, focused on children's safety
Money app & debit card with gamified learning.	Ai-powered identity theft & fraud prevention.	Non-profit digital safety & online wellbeing resources.
UK	**U.S.A**	**UK**

We want to combine all three of these functions, and link social media behaviour directly to financial risk.

3.

Stakeholders

Primary and Secondary stakeholders for our project

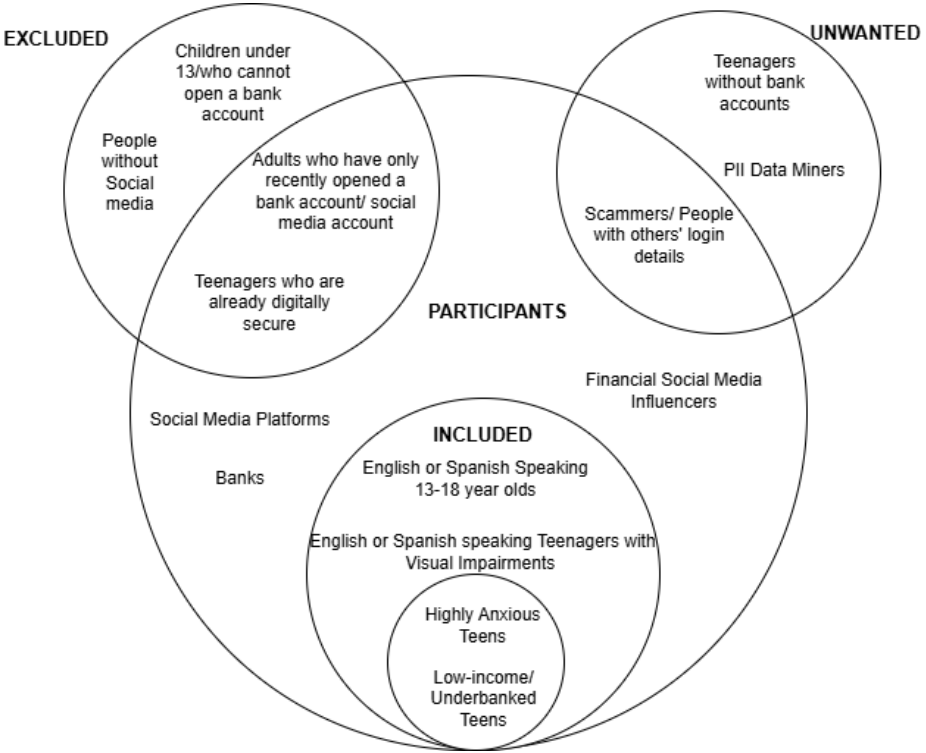
Stakeholders & Inclusivity

Primary

- UK-Based 13–18-year-olds with social media and bank accounts
- Spanish speaking 13–18-year-olds
- Visually Impaired

Secondary

- UK-Based Banks
- Social Media Platforms
- Financial Influencers



Panda Inclusivity Map

4.

System Overview

Explanation of our solution

Dylan

Solution Breakdown

Web Application

Maximise engagement without setup

API

To authenticate users and collect posts

Machine Learning

To identify posts containing PII

Education

To show & explain financially risky social media behaviour

*Our solution is a **web app** that uses **NLP** to identify personal identifying information leaked in social media posts and highlight it for users, helping UK teens understand and **mitigate digital and financial security risks**.*

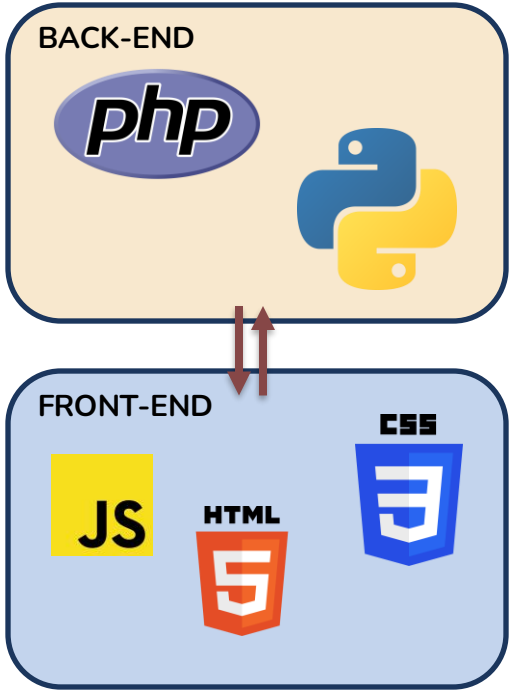
Solution Technologies

Back-end

- **PHP:**
  → Authentication of user's social media accounts.
  → Runs the NLP Python script on posts.
  → Parses results for the results page
- **Python:**
  → Runs NLP model against user posts, identifying PII Information if present.

Front-end

- **Javascript:**
  → Renders severity level tags and pie chart.
  → Renders findings summary for the user.
  → Handles displaying error codes
- **CSS & HTML:**
  → HTML defined the structure of the website (such as navbar, results, header, footer etc).
  → CSS used to stylise HTML elements.



DEMO

Natural Language Processing

NLP Implementation (1/2)

- Uses Named Entity Recognition (NER) → categorisation of information in text (IBM 2023).
- Detects PII within social media posts.
- NER is facilitated using the GLiNER Python Library.
  - Lightweight.
  - Allows for detection using **custom** entity labels.
- GLiNER already have their own general pre-trained data sets with a range of sizes on Hugging Face:
  - These datasets → are unreliable.
- Nvidia has their own PII dataset -> Reliable! (covered in testing section)

```python
# post_history is a string passed in which contains a single historic post
post_history = sys.argv[1]

# Disable symlinks on Windows - causes permissions issues if not!!
os.environ["HF_HUB_DISABLE_SYMLINKS"] = "1"

# Squash output
logging.disable_progress_bar()
warnings.filterwarnings("ignore")

# get pretrained model from hugging face which is compatible with gLiNER
model = GLiNER.from_pretrained("nvidia/gliner-PII")

PPI_counts = {
  "FIRST_NAME": 0,
  "SURNAME": 0,
  "DATE_OF_BIRTH": 0,
  "MOBILE_NUMBER": 0,
  "NATIONAL_INSURANCE_NUMBER": 0,
  "SCHOOL_NAME": 0,
  "BIRTHPLACE": 0,
  "ADDRESS": 0,
  "BANK_ACCOUNT_NUMBER": 0,
  "SORT_CODE": 0,
  "BACS": 0,
  "IBAN": 0,
  "DEBIT_CARD_NUMBER": 0,
  "CARD_CVC": 0,
  "CARD_EXPIRY": 0
}

labels = list(PPI_counts.keys())
```

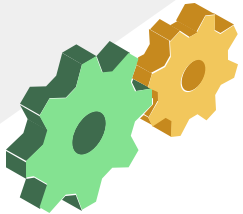
NLP Implementation (2/2)

- **Input** - a single user social media post (string)
- **For each post** – Runs against pretrained model using PII dataset and adds to label count.
- **For each label** - iterate and turn into human-readable format.

Example:
"Why isn't my bank working today!
I've tried everything. I'm using 748576489 with 00-24-12.
Am I missing something, or
is my bank just testing my patience today?"
(Anthropic 2023)

Will return: *"BANK ACCOUNT NUMBER, AND SORT CODE"*

```python
""" Runs NER analysis on histroic user posts
This function runs NER analysis on historic user posts. It will then add a count
to the PII count object defined above, which is then parsed to create a human-readable
string i.e. "Cvv, Iban, and Sort Code"

"""
def NER_anaysis() -> str:
  flags = []

  entities = model.predict_entities(post_history, labels)
  for ent in entities:
    PPI_counts[ent["label"]] += 1

  for key, value in PPI_counts.items():
    if value > 0:
      flags.append(key.replace("_", " ").upper())

  flags_length = len(flags)

  if flags_length == 0:
    return "None"
  elif flags_length == 1:
    return flags[0]
  else:
    return (", ".join(flags[:-1]) + f", and {flags[-1]}")

if __name__ == "__main__":
    results = NER_anaysis();
    print(results)
```



ACCESSIBILITY

Hritika

Clear Design & Navigation

- Following Web Content Accessibility Guidelines 2.2- focusing on colour & contrast
- Aiming for AA compliancy- matching the UK Government
- Colours such as Gold and Coral (#C6891E & #E75757) only to be used in large text, icons and buttons.
- Screen reader compatible in Spanish & English
- Purposeful Alt-Text & Naming consistency

Light	Dark
Hex #FFFFFF	Hex #1B365D
Hex #C6891E	Hex #F2C75C
Hex #3E6B4D	Hex #84E291
Hex #991B1B	Hex #E75757
Hex #667A81	Hex #D0D3D4
Hex #1B365D	Hex #FFFFFF

6.

Impact

What is the impact and implications of our solution?

Impact & Limitations

Social	Ethical	Accessibility
Financial awareness and digital education for long-term social mobility	We teach youths that their PII is valuable .	WCAG 2.2 1.4.3 Contrast compliant
Impact is limited to a willingly educated youth.	To stay ethical, we cannot store data beyond the session.	Voice navigable in Spanish and English
Potential 'anxiety factor'.	Must ensure using social spiders will not influence labelling of a 'high-risk' person	Reliant on device screen readers and voice access instead of built- in functionality

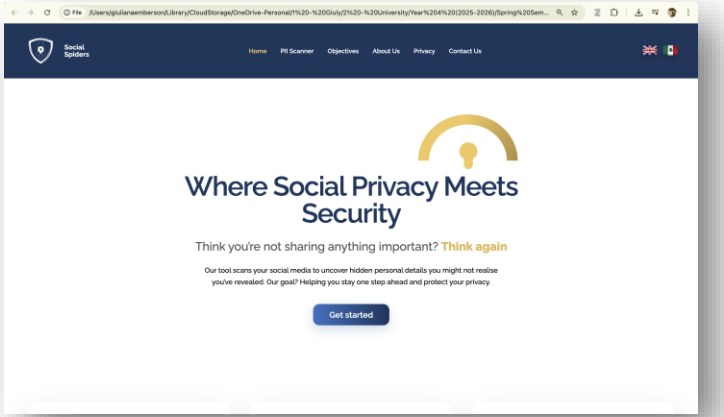
7.

Testing

1. Account authentication works reliably.
2. NLP correctly detects personal identifiable information.
3. The website functions in both English and Spanish.
4. Accessibility standards are met.
5. API integrations work.
6. The system handles edge cases and fails gracefully.
7. The full user experience works and is easy to use.

Front-End Testing

Qualitative testing



Localisation Testing

1. Website works in both English and Spanish
2. The toggle to switch languages works as expected
3. All UI text is translated
4. Switching languages mid-session works
5. The NLP model still works as expected

Front-End Testing

Qualitative testing

Design vs Actual UI Experience

- Light Mode works as expected. We implemented the design accurately and added more information about privacy and data collection
- Dark mode to be added in future

Front-End Testing

Qualitative testing

Test Account
A dummy account with posts to cover different cases:

1. Multiple posts with PII
2. Phone number, email and credit card number
3. Testing a Spanish phone number within Spanish Text
4. Posts with special characters

Most of these tests passed well, but emails were often missed.

Front-End Testing

User testing

1. User visits site
2. User selects language using toggle (English/Spanish)
3. User navigates to the PII scanner
4. User starts a scan
5. User logs into their Facebook account and grants permissions
6. The NLP analyses the user's posts
7. The results of the NLP analysis is displayed to the user

We would like to test with more users before the product is ready

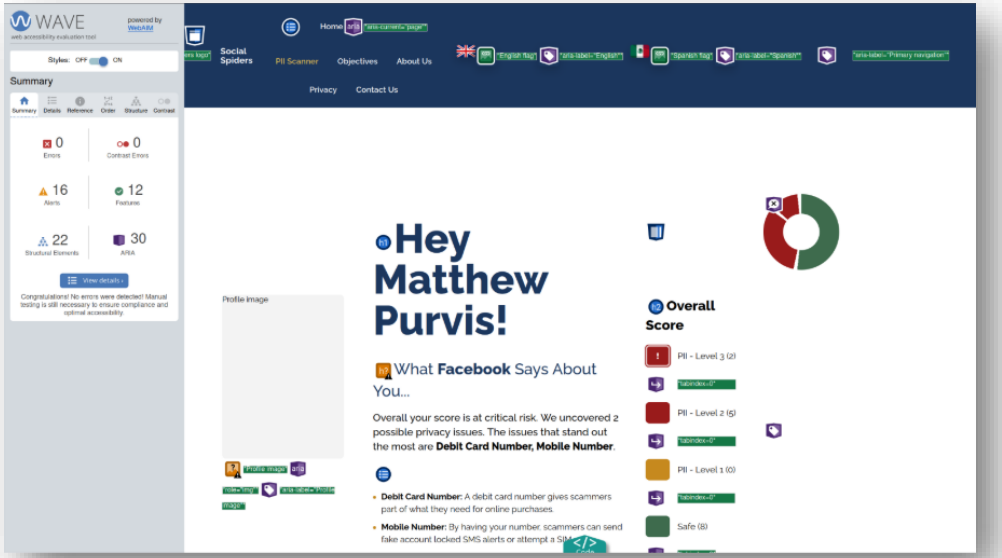
Front-End Testing

Qualitative testing

Accessibility testing using WAVE Web Accessibility Evaluation Tools

- Ensures compatibility with WCAG AA:
  - → Aria Labels & Alt tags
  - → Colour contrast
  - → Order of screen reader

- Screen reader works well



Back-End Testing

Quantitative Testing

Unit testing of the NLP model

- Testing the script works as expected
- No entity, single entity, multiple, duplicate and an unknown label

```
-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
========================================================== 5 passed, 9 warnings in 8.85s ==========================================================
```

NLP model performance

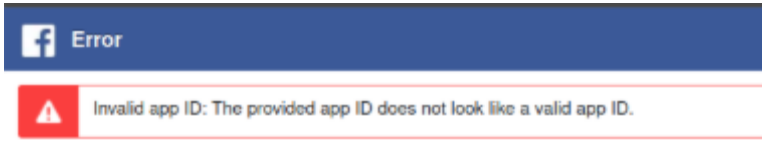
- Generated a dataset that included English, social media style language, Spanish and edge cases

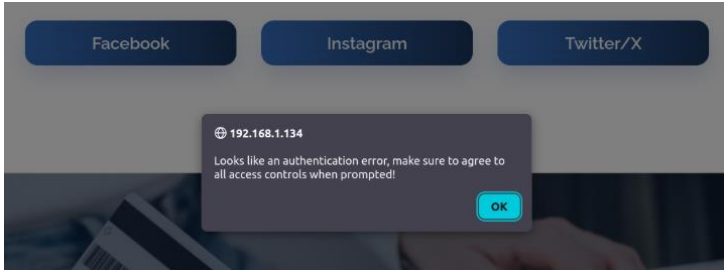
```
eleanor@eleanorsair329 src % python3 evaluate_nlp.py
Accuracy: 0.82
Precision: 0.83
Recall: 0.97
F1 Score: 0.89
```

Error Testing

Error handling:

- Errors such as an invalid API key are handled by the Social Media sites
- User fails to authenticate
- Can’t find the posts
- User hasn’t selected a platform





Insights & Patterns

- We have tested against all our objectives
- The product is sturdy and handles errors well
- We have chosen a suitable NLP model
- The UI is accessible and appealing
- In future, we will set up a full test pipeline and handle more errors

8.

Conclusion & Future Work

Conclusion

By the time our users turn 18, they won't just have a bank account - they will have a high Security Score and a clean digital footprint, making them the most 'low-risk' customers in the bank's portfolio.

Ellie

Future Work

More Social Media support

X, Instagram, and more.

Visible Posts: Grid-view

Show the user the exact posts leaking information.

PDF Report Sharing

Allows the user to share and save their results.

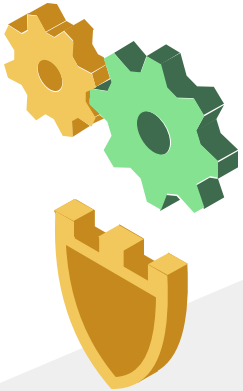
Dark Mode

Move the dark-mode from prototype to 'live'

THANK YOU

Any Questions?

CREDITS: This presentation template was created by **Slidesgo,** and includes icons by **Flaticon** and infographics & images by **Freepik**



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